



Review

Quadriceps tendon versus hamstring tendon graft for primary anterior cruciate ligament reconstruction: A systematic review and meta-analysis of randomised trials



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ARTICLE INFO

Article history:

Received 6 March 2024

Revised 13 May 2024

Accepted 2 July 2024

Keywords:

ACL reconstruction
Quadriceps tendon
Hamstring tendon
Autograft
Systematic review
Meta-analysis

ABSTRACT

Background: Anterior cruciate ligament reconstruction (ACLR) is most commonly performed with hamstring tendon (HT) or bone-patellar tendon-bone (BTB) autografts, although the quadriceps tendon (QT) autograft has recently increased in popularity. This systematic review and meta-analysis review compares QT and HT autografts for primary ACLR with a sole focus on randomised controlled trials (RCTs).

Methods: A prospective protocol was registered on PROSPERO (CRD42023427339). The search included MEDLINE, Embase and Web of Science until February 2024. Only comparative RCTs were included. The primary outcome was the International Knee Documentation Committee (IKDC) Subjective Knee Evaluation Form score. Secondary outcomes included: other validated patient-reported outcome measures (PROMs), objective strength scores, complications, and return to sport and work.

Results: From 2,609 articles identified, seven were included ($n = 474$ patients). This meta-analysis did not identify a significant difference in post-operative IKDC scores (5 articles; $p = 0.73$), Lysholm scores (3 studies; $p = 0.80$) or Tegner activity scales (2 studies; $p = 0.98$). There were no differences in graft failure rates (4 studies; $p = 0.92$) or in overall adverse events (4 studies; $p = 0.83$) at 24 months post-ACLR as per meta-analysis. Donor site morbidity scores were significantly lower in the QT group (MD -4.67 , 95% CI -9.29 to -0.05 ; 2 studies, 211 patients; $p = 0.05$, $I^2 = 34\%$).

Conclusion: There were no differences between QT and HT in PROMs, graft failure rates or overall complications based on low- to moderate-quality evidence. There may possibly be lower donor site morbidity with the QT autograft, however, the evidence is not sufficient to draw definitive conclusions.

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Abbreviations: ACL, Anterior cruciate ligament; ACLR, Anterior cruciate ligament reconstruction; BTB, Bone-patellar tendon-bone; CI, Confidence interval; GRADE, Grading of Recommendations, Assessment, Development, and Evaluations; HT, Hamstring tendon; IKDC, International Knee Documentation Committee (Subjective Knee Evaluation Form score); KOOS, Knee Injury and Osteoarthritis Outcome Score; LSI, Limb symmetry index; MCID, Minimal clinically important difference; MD, Mean difference; NLR, National Ligament registry; OR, Odds ratio; PROMs, Patient-reported outcome measures; QT, Quadriceps tendon; RCT, Randomised controlled trial; RoB 2.0, Risk of bias 2.0; VAS, Visual analogue scale.

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<https://doi.org/10.1016/j.knee.2024.07.002>

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1. Introduction

In a population-based study conducted in the United States, the age- and sex-adjusted annual incidence of anterior cruciate ligament (ACL) rupture was 68.6 per 100,000 person-years, making this one of the most prevalent orthopaedic injuries [1]. An ACL rupture can have a direct negative impact on quality of life and activity levels while also increasing the risk of developing secondary osteoarthritis in the knee [2]. In the United Kingdom, the rate of ACL reconstruction (ACLR) has increased 12-fold between 1997 and 2017 to 24.2 ACLR operations per 100,000 individuals [3]. A recent pragmatic randomised controlled trial in the United Kingdom (UK) revealed that ACLR was clinically superior and more cost-effective than rehabilitation alone in patients with non-acute ACL injuries [4].

According to the UK National Ligament Registry (NLR) annual report in 2022, autografts were utilised in 98% of primary ACLRs, of which the hamstring tendon (HT) graft was used in 91% of cases followed by the bone-patellar tendon-bone (BTB) graft in 8% and the quadriceps tendon (QT) autograft in approximately 0.5% [5]. A cross-sectional survey by the international ACL Study Group revealed that 50% of members use a HT autograft as their primary choice for ACLR, followed by 40% of members opting for BTB as their preferred graft choice [6]. The quadriceps tendon (QT) autograft has risen in popularity over the last decade and is now the primary choice for 9% of surgeons in the study group [6].

With the current variations in graft choice and use, it has been suggested that further investigations on patient-specific graft selection is required to determine the ideal graft choice for patients undergoing ACLR [7]. HT and BTB autografts have been compared extensively in the literature [8], while BTB and QT autografts are currently being compared as part of the STABILITY 2 trial in patients at high risk of graft failure [9]. Currently, there is no clear consensus in the literature when comparing HT and QT grafts.

A systematic review of the QT autograft in ACLR by Hurley et al. in 2018 reported that QT is a viable autograft option with “comparable knee stability, functional outcomes, donor-site morbidity, and re-rupture rates” when compared to HT and BTB autografts, however, a *meta-analysis* was not performed (nine retrospective, five prospective and one randomised; 15 studies, $n = 1,910$) [10]. The most recent systematic review that compared QT and HT autografts by Tan et al. included 13 retrospective, 3 prospective and 6 randomised studies until November 2020 (22 studies, $n = 28,621$) [11]. Their review found comparable graft failure rates, patient-reported outcome measures (PROMs), clinical outcomes and functional outcomes, however, the authors note that some of the results of the *meta-analysis* showed high levels of heterogeneity and that the mixing of randomised controlled trials and observational studies may have skewed the results [11]. Therefore, this system-

atic review and meta-analysis aims to provide an up-to-date comparison of QT and HT autografts for primary anterior cruciate ligament reconstruction using only randomised controlled trials.

2. Methods

2.1. Search strategy

A protocol for this systematic review was prospectively published on PROSPERO (registration ID CRD42023427339). This review is reported as per the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) criteria [12,13]. The search strategy has been included (Appendix A). The PRISMA checklist has also been included (Appendix B). MEDLINE, Embase and Web of Science were systematically searched for eligible articles up to 1 February 2024.

Articles identified from the database searches were screened by title and abstract by two authors (SR and AR). Thereafter, the full manuscript of the final articles was assessed against eligibility criteria by two independent authors (SR and AR). Any dispute was discussed by all authors and settled by a consensus. Data from eligible articles were inputted into a pre-defined, piloted spreadsheet that was reviewed by an additional author (HKCS).

2.2. Eligible studies and participants

Only randomised controlled trials (RCTs) or quasi-randomised studies that compare quadriceps tendon autografts with hamstring tendon autografts for primary anterior cruciate ligament reconstruction were included. Articles of any language were included provided that an English translation was available at the time of the search.

Studies with the following were excluded: anterior cruciate ligament repair; anterior cruciate ligament reconstruction with any allograft; anterior cruciate ligament revision or secondary anterior cruciate ligament reconstruction; meniscal pathology requiring immediate surgery; and inflammatory or degenerative knee pathology.

2.3. Eligible interventions and comparators

The eligible intervention was the quadriceps tendon (QT) autograft (with or without a bone block) utilised for primary anterior cruciate ligament reconstruction. The eligible comparator was any hamstring tendon (HT) autograft for primary anterior cruciate ligament reconstruction regardless of whether this was single-bundle, double-bundle or quadruple-bundle autografts. Reconstructions with or without a lateral extra-articular tenodesis (LET) were eligible for inclusion.

2.4. Outcome measures

The primary outcome was the International Knee Documentation Committee (IKDC) Subjective Knee Evaluation Form score at final follow-up. The IKDC is a patient-reported outcome measure (PROM) that reports patients' knee symptoms, functional status and activities with lower scores representing greater severity of symptoms and decreased levels of function [14].

Secondary outcomes were divided into other validated PROMs, objective strength scores, complications, and time taken to return to previous activities. Validated PROMs included: the ACL-Quality of Life (QOL) [15], Knee injury and Osteoarthritis Outcome Score (KOOS) [16], Lysholm Knee Questionnaire [17], Tegner Activity Scale [18], Cincinnati Knee Rating System (CKRS) [19], and visual analogue scale (VAS) for pain scores [20]. Studies were also screened for isokinetic and isometric hamstring strength and quadricep strength. Complications were categorised into harvest-site infection rate, donor site morbidity, graft failure rate and overall adverse events, which included any other negative outcomes, such as quadriceps or hamstrings atrophy. Time taken to return to work and return to sport were also recorded.

The minimal clinically important differences (MCIDs) for the chosen outcomes were searched in the literature and recorded to aid the interpretation of the analysis. The MCID was between 9.0 and 9.5 for IKDC and between 10.0 and 10.1 for Lysholm scores at two years and as reported by two studies investigating MCID after ACLR [21,22]. Another study recorded the MCID for Tegner scores as 1.0 for patients with ACL injuries [23].

2.5. Assessment of risk of bias

The Cochrane Risk of Bias 2.0 tool was used to assess risk of bias across all of the randomised trials included in this review [24]. The quality of evidence and strength of recommendations were assessed via the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) framework [25]. Two authors (AR and HKCS) independently assessed risk of bias of included studies and assessed the quality of evidence as per GRADE. Another author (CK or IA) was consulted if there were any discrepancies.

2.6. Data analysis

QT and HT autografts were compared via a narrative synthesis. The quantitative data for primary and secondary outcomes that were available have been presented in separate tables. Continuous variables were presented as mean or median with standard deviation or interquartile range. Categorical variables have been reported as percentages.

A quantitative *meta-analysis* was conducted on the Review Management (RevMan) software to compare outcomes between QT and HT autografts [26]. Outcomes at final follow-up were pooled when conducting the *meta-analysis* with an intention-to-treat analysis and a random effects model. When applicable, mean differences and odds ratios were calculated with confidence intervals provided. A p-value threshold of 0.05 was used when determining statistical significance. Studies that presented data with disparate or incomparable outcomes were not included in the quantitative analysis and were instead discussed in the narrative synthesis. A discussion of possible explanations and an overall summation of the findings of this review have been presented in the discussion and conclusion sections, respectively.

3. Results

3.1. Study selection

In total, 2,609 articles were identified after systematically searching the relevant databases: Embase, MEDLINE and Web of Science. After removing 951 duplicates and screening abstracts, 45 full-text articles were assessed for eligibility. Of these, only 7 articles were eligible for inclusion (Figure 1) [27–33].

3.2. Study characteristics

The baseline characteristics of the seven included articles have been outlined (Table 1) [27–33]. The studies were conducted across six countries on three continents between 2010 and 2020. One study, which was published in 2020, did not report its recruitment period [31]. Two articles had a follow-up of 12 months [30,31] while four articles had a follow-up of 24 months [27–29,32]. One study had a follow-up of 48 hours as the outcomes of interest were post-operative pain scores and frequency of analgesia consumption [33].

In total, there were 237 patients in the QT group and 237 patients in the HT group [27–33]. Four of the articles described utilising QT grafts with patellar bone block and HT as a 4-strand graft [27–30], but three articles did not specify the exact technique used [31–33]. The mean age and standard deviation of participants ranged from 18.7 ± 3.6 to 29.2 ± 8.5 in the

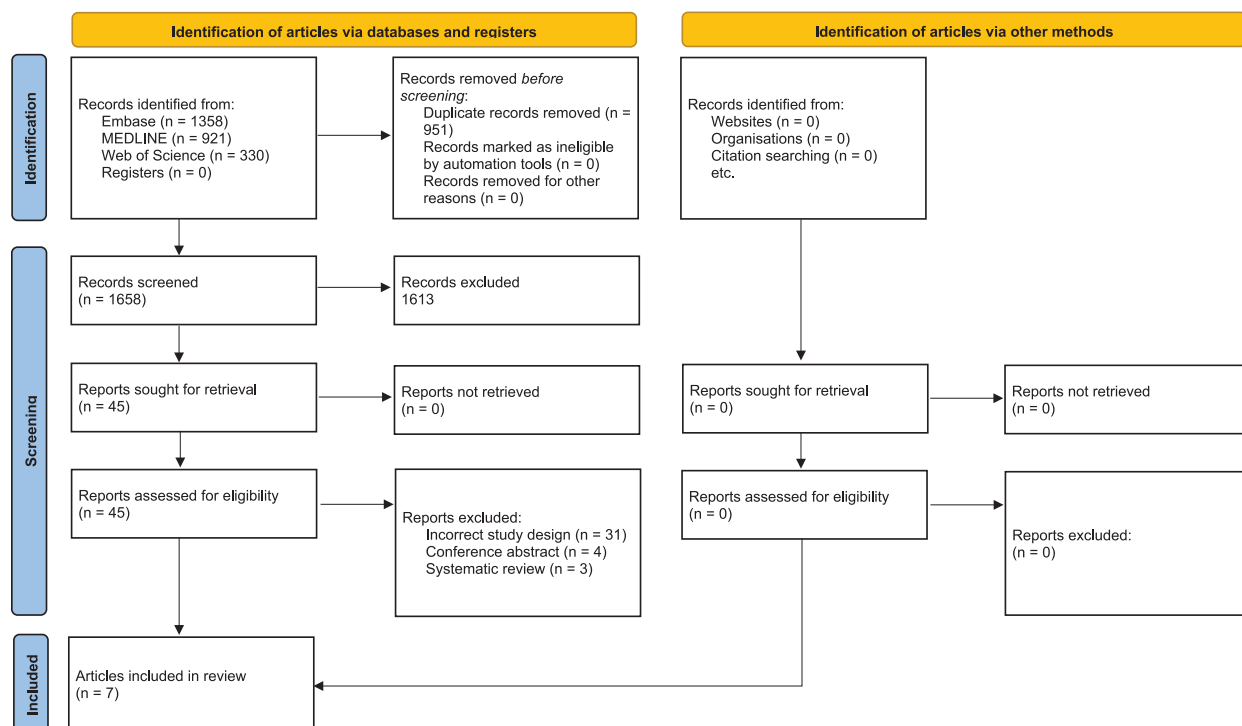


Figure 1. PRISMA 2020 flow diagram of identified, screened and included articles.

Table 1
Baseline characteristics of all included studies.

Study ID (Author, country and year of publication)	Type of study	Recruitment Period (Years)	Technique		Number of patients		Mean age (SD)		Mean follow-up duration (months)		Patient Reported Outcome Measures	
			QT	HT	QT (M/F)	HT (M/F)	QT	HT	QT	HT		
Ebert et al., Australia (2024)	RCT Prospective	March 2019 to December 2020	QT with patellar bone block	HT as 4-strand graft	57 (28/29)	55 (28/27)	28.1 (8.2)	29.4 (7.7)	24	24	IKDC Score Lysholm Questionnaire Tegner Activity Scale Cincinnati Knee Rating System VAS for pain	Knee
Horstmann et al., Germany (2022)	RCT Prospective	December 2010 to June 2013	QT with patellar bone block	HT as 4-strand graft	24 (21/3)	27 (12/15)	24.1 (3.6)	32.7 (11.4)	24	24	IKDC Score Lysholm Questionnaire	Knee
Lind et al., Denmark (2020)	RCT Prospective	May 2014 to January 2016	QT with patellar bone block	HT as 4-strand graft	50 (29/21)	49 (25/24)	27.2 (6.4)	27.1 (6.1)	24	24	IKDC Score KOOS Tegner Activity Scale	
Sinding et al., Denmark (2020)	RCT Prospective	May 2014 to January 2016	QT with patellar bone block	HT as 4-strand graft	42 (25/17)	43 (23/20)	28.3 (6.2)	28.7 (6.4)	12	12	IKDC Score	
Vilchez-Cavazos et al., Mexico (2020)	RCT Prospective	NR	NS	NS	14 (11/3)	14 (12/2)	23.0 (7.42)	23.0 (8.16)	12	12	IKDC score Lysholm Questionnaire VAS for pain	Knee
Martin-Alguacil et al., Spain (2018)	RCT Prospective	July 2015 to July 2016	NS	NS	26 (23/3)	25 (16/9)	18.7 (3.6)	19.2 (3.6)	24	24	Lysholm Questionnaire Cincinnati Knee Rating System	Knee
Buescu et al., Romania (2017)	RCT Prospective	October 2013 to May 2015	NS	NS	24 (NR/NR)	24 (NR/NR)	29.2 (8.5)	27.5 (5.6)	0 (48 h)	0 (48 h)	VAS for pain	

QT = Quadriceps tendon (QT) graft; HT = Hamstring tendon (HT) graft.
(M/F) = (male/female).
RCT = randomised controlled trial; SD = standard deviation
NR = Not reported; NS = Not specified.

QT group and 19.2 ± 3.6 to 32.7 ± 11.4 in the HT group [27–33]. Two of the published articles were part of the same trial: Lind et al. reported PROMs at 24 months [29] and Sinding et al. focused on muscle strength and functional capacity at 12 months [30].

3.3. Primary outcome: IKDC

Five articles reported post-operative IKDC scores (Table 2) [27–31], of which three articles reported scores at 24 months [27–29] and two articles reported scores at 12 months [30,31]. Pooled analysis did not reveal a statistically significant difference between the groups in terms of post-operative IKDC scores (MD 0.52, 95% CI -2.40 to 3.43; 5 articles, 375 patients; $p = 0.73$, Figure 2). There was low heterogeneity ($I^2 = 40\%$) and the quality of evidence was moderate as per the GRADE rating system (Table 3). The mean difference was less than the MCID of 9.0–9.5 and is therefore unlikely to be of clinical importance [21,22].

3.4. Secondary outcomes: Patient-reported outcome measures

The relevant PROMs have been outlined in Table 2 [27–33]. One article did not report any of the PROMs in this section [30]. None of the included articles reported ACL-QoL scores [27–33]. Only one study reported KOOS scores, which found no statistically significant difference in KOOS subscales between QT and HT groups 24 months post-ACLR [29].

Four studies reported Lysholm scores, none of which reported a significant difference between QT and HT groups [27,28,31,32]. Only three of these were eligible for pooled analysis [27,28,31], which revealed a non-significant mean difference of 0.55 in favour of HT (MD -0.55, 95% CI -4.88 to 3.77; 3 studies, 191 patients; $p = 0.80$, Figure 3.1). There was moderate heterogeneity ($I^2 = 56\%$) and the quality of evidence was deemed to be low as per GRADE (Table 3). The mean difference was less than the MCID of 10.0 to 10.1 and is therefore unlikely to be of clinical importance [21,22].

Two studies reported Tegner activity scales [27,29], and pooled analysis did not reveal a difference at 24 months post-ACLR (MD 0.00, 95% CI -0.39 to 0.40; 2 studies, 211 patients, $p = 0.98$, Figure 3.2). There was low heterogeneity ($I^2 = 0\%$) and quality of evidence was moderate (Table 3). The mean difference was less than the MCID of 1.0 [23], indicating that this is unlikely to be a clinically important difference.

Only one of the articles reported CKRS scores with no difference between QT and HT groups at 24 months post-ACLR [27]. The same study reported no differences in VAS scores for knee pain and graft-site pain between QT and HT groups at 24 months post-ACLR [27]. Another study reported a non-significant difference in VAS between the groups at 12 months [31]. One study with a follow-up of 48 hours found that patients in the QT group reported less pain and need for analgesia in the immediate post-operative period [33]. This outcome was not suitable for meta-analysis due to the variability in outcome reporting with respect to site of pain and time relative to surgery.

3.5. Secondary outcomes: Objective strength scores

Only two studies reported objective strength scores [27,30]. Ebert et al. found that patients in the HT group had significantly greater quadriceps strength limb symmetry indices (LSIs) at 6 and 12 months ($p < 0.05$), while the QT group had significantly greater hamstring strength LSIs at 6, 12 and 24 months post-ACLR [27]. Sinding et al. found that the HT autograft led to moderate impairments of knee extensor (KE) and flexor (KF) muscle strength while the QT autograft affected KE only, although in a more pronounced manner [30]. Due to the disparity of reporting strength outcomes, meta-analysis was not performed.

3.6. Secondary outcomes: Complications

All complications that were reported in the included studies are outlined in Table 2 [27–29,32]. Only Horstmann et al. reported harvest site infection rates, which were not significantly different between QT and HT ($p > 0.05$) [28].

None of the studies reported overall donor site morbidity rates. However, two studies reported mean donor site morbidity scores [27,29]. Meta-analysis revealed a lower donor site morbidity score in the QT group at 24 months, which was statistically significant (MD -4.67, 95% CI -9.29 to -0.05; 2 studies, 211 patients; $p = 0.05$, Figure 4), however, the MCID for this outcome is not known. There was low heterogeneity ($I^2 = 34\%$) and the quality of evidence was moderate (Table 3). Lind et al. also reported that the QT group had a relatively lower donor site morbidity score at 12-months ($p = 0.03$) [29].

Four studies reported graft failure rates at 24 months [27–29,32], and a pooled analysis did not reveal a difference between the 6 failures in the QT group and 6 failures in the HT group (OR 1.06, 95% CI 0.32 to 3.53; 4 studies, 313 patients; $p = 0.87$, $I^2 = 0\%$; Figure 5.1) with moderate quality of evidence (Table 3).

Four studies reported overall adverse events at 24 months and were eligible for meta-analysis [27–29,32], which did not find a significant difference (OR 1.08, 95% CI 0.51 to 2.28; 4 studies, 313 patients; $p = 0.83$, $I^2 = 0\%$; Figure 5.2) with moderate quality of evidence (Table 3). None of the studies reported quadriceps or hamstring atrophy [27–33].

Table 2

Primary and secondary outcomes: IKDC, other PROMs and complications.

Primary outcome: IKDC								
Study		Ebert et al., 2024	Horstmann et al., 2022	Lind et al., 2020	Sinding et al., 2020	Vilchez-Cavazos et al., 2020	Martin-Alguacil et al., 2018	Buescu et al., 2017
Pre-operative IDKC score(SD)	QT	49.7 (17.9)	66.8 (16.9)	NR	NR	57.0 (15.9)	NR	NR
	HT	50.6 (18.9)	59.0 (17.2)	NR	NR	57.0 (17.0)	NR	NR
	p-value	NR	0.109	-	-	NR	-	-
Post-operative IKDC score (SD)	QT	89.1 (9.6)	89.3 (12.2)	82 (14)	76 (17)	90.0 (6.29)	NR	NR
	HT	91.7 (7.5)	83.7 (12.7)	78 (18)	76 (15)	90.0 (4.45)	NR	NR
	p-value	0.447	0.169	0.25	>0.05	0.505	-	-
Secondary outcomes: other PROMs								
Study		Ebert et al., 2024	Horstmann et al., 2022	Lind et al., 2020	Sinding et al., 2020	Vilchez-Cavazos et al., 2020	Martin-Alguacil et al., 2018	Buescu et al., 2017
ACL-QOL(SD)	QT	NR	NR	NR	NR	NR	NR	NR
	HT	NR	NR	NR	NR	NR	NR	NR
	p-value	-	-	-	-	-	-	-
KOOS - Symptoms(SD)	QT	NR	NR	82 (16)	NR	NR	NR	NR
	HT	NR	NR	79 (19)	NR	NR	NR	NR
	p-value	-	-	0.34	-	-	-	-
KOOS - Pain(SD)	QT	NR	NR	87 (12)	NR	NR	NR	NR
	HT	NR	NR	88 (13)	NR	NR	NR	NR
	p-value	-	-	0.49	-	-	-	-
KOOS - ADL(SD)	QT	NR	NR	91 (11)	NR	NR	NR	NR
	HT	NR	NR	95 (10)	NR	NR	NR	NR
	p-value	-	-	0.09	-	-	-	-
KOOS - Sport and Recreation (SD)	QT	NR	NR	70 (23)	NR	NR	NR	NR
	HT	NR	NR	76 (16)	NR	NR	NR	NR
	p-value	-	-	0.16	-	-	-	-
KOOS - QOL(SD)	QT	NR	NR	60 (18)	NR	NR	NR	NR
	HT	NR	NR	63 (20)	NR	NR	NR	NR
	p-value	-	-	0.46	-	-	-	-
Lysholm(SD)	QT	92.5 (8.0)	90.4 (11.9)	NR	NR	95.0 (6.67)	93.8 (NR)	NR
	HT	94.4 (6.6)	83.5 (17.4)	NR	NR	98.0 (7.42)	92.7 (NR)	NR
	p-value	0.206	0.131	-	-	0.828	0.892	-
Tegner Scale(SD)	QT	6.7 (1.4)	NR	6.0 (1.5)	NR	NR	NR	NR
	HT	6.8 (1.7)	NR	5.9 (1.3)	NR	NR	NR	NR
	p-value	0.679	-	0.71	-	-	-	-
CKRS(SD)	QT	92.8 (8.7)	NR	NR	NR	NR	NR	NR
	HT	94.3 (7.2)	NR	NR	NR	NR	NR	NR
	p-value	0.391	-	-	-	-	-	-
VAS(SD)	QT	0.9 (1.2)	NR	NR	NR	0.0 (IQR 0.0 1.5)	NR	4.00 (IQR 3.25–4.25)
	HT	0.7 (0.8)	NR	NR	NR	0.0 (IQR 0.0 1.0)	NR	5.00 (IQR 4.00–6.00)
	p-value	0.776	-	-	-	0.450	-	NR

Table 2 (continued)

Secondary outcomes: complications								
Study		Ebert et al., 2024	Horstmann et al., 2022	Lind et al., 2020	Sinding et al., 2020	Vilchez-Cavazos et al., 2020	Martin-Alguacil et al., 2018	Buescu et al., 2017
Graft failure rate (n/total)	QT	1.75 (1/57)	12.50 (3/24)	2.00 (1/50)	NR	NR	3.85 (1/26)	NR
	HT	0.00 (0/55)	7.40 (2/27)	2.04 (1/49)	NR	NR	12.00 (3/25)	NR
	p-value	NR	>0.05	>0.05	-	-	0.10	-
Harvest site infection rate (n/total)	QT	NR	4.17 (1/24)	NR	NR	NR	NR	NR
	HT	NR	0.00 (0/27)	NR	NR	NR	NR	NR
	p-value	-	>0.05	-	-	-	-	-
Donor site morbidity rate (n/total)	QT	NR	NR	NR	NR	NR	NR	NR
	HT	NR	NR	NR	NR	NR	NR	NR
	p-value	-	-	-	-	-	-	-
Donor site morbidity score (SD)	QT	9.3 (10.6)	NR	14 (17)	NR	NR	NR	NR
	HT	12.3 (10.7)	NR	22 (18)	NR	NR	NR	NR
	p-value	0.301	-	*0.046	-	-	-	-
Adverse events rate (n/total)	QT	12.28 (7/57)	16.67 (4/24)	10.00 (5/50)	NR	NR	3.85 (1/26)	NR
	HT	9.09 (5/55)	7.40 (2/27)	12.24 (6/49)	NR	NR	12.00 (3/25)	NR
	p-value	NR	>0.05	>0.05	-	-	0.10	-

IKDC = International Knee Documentation Committee (IKDC) Subjective Knee Evaluation Form; ACL-QoL = ACL-Quality of Life (QOL); Knee Injury and Osteoarthritis Outcome Score; KOOS - ADL = KOOS - Activities of daily living section; KOOS - QOL = KOOS - Quality of Life section; Lysholm = Lysholm Knee Questionnaire Score; Tegner = Tegner Activity Scale;
CKRS = Cincinnati Knee Rating System, VAS = Visual Analogue Score for Pain.
QT = Quadriceps tendon (QT) graft. HT = Hamstring tendon (HT) graft.
SD = standard deviation. NR = Not reported.
*Indicates statistical significance i.e. p-value < 0.05.

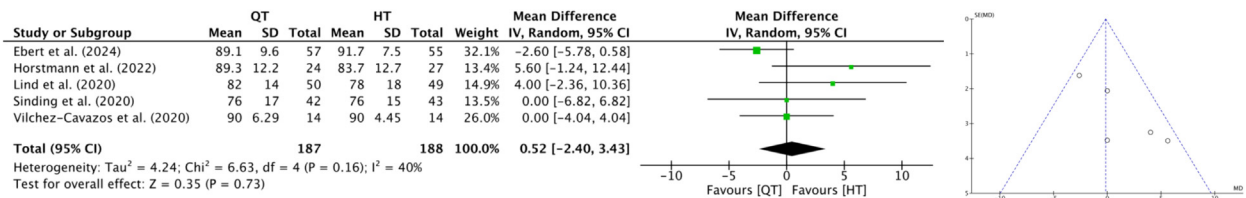
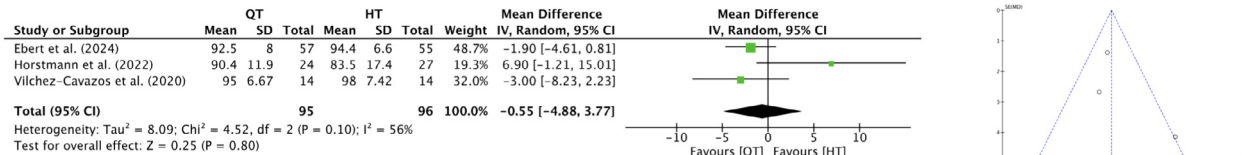


Figure 2. Meta-analysis of post-operative IKDC scores at final follow-up with an associated forest plot.

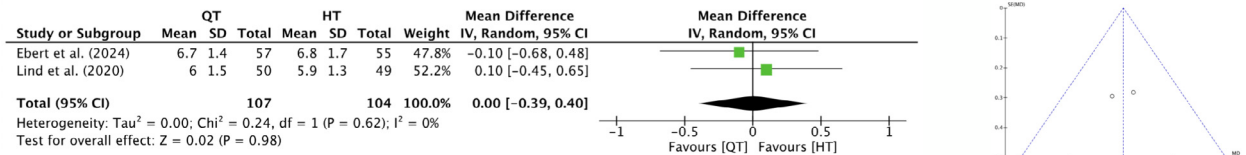
Table 3
Quality of evidence of outcomes as assessed by the GRADE framework.

Outcomes	Number of studies (number of patients)	Risk of bias	Imprecision	Inconsistency	Indirectness	Publication bias	Overall GRADE Rating
IKDC	5 (375)	Moderate	Low	Low	Low	Low	Moderate
Lysholm	3 (191)	Moderate	Moderate	Moderate	Low	Low	Low
Tegner activity scale	2 (211)	Moderate	Low	Low	Low	Low	Moderate
Donor site morbidity score	2 (211)	Moderate	Low	Low	Low	Low	Moderate
Adverse event rate	4 (313)	Moderate	Low	Low	Low	Low	Moderate
Graft failure rate	4 (313)	Moderate	Low	Low	Low	Low	Moderate

IKDC = International Knee Documentation Committee (IKDC) Subjective Knee Evaluation Form; Lysholm = Lysholm Knee Questionnaire Score; Tegner = Tegner Activity Scale;
GRADE = Grading of Recommendations, Assessment, Development, and Evaluations.



3.1 - Lysholm scores



3.2 - Tegner activity scale

Figure 3. Meta-analysis of post-operative 1) Lysholm scores and 2) Tegner activity scale at final follow-up with associated forest plots.

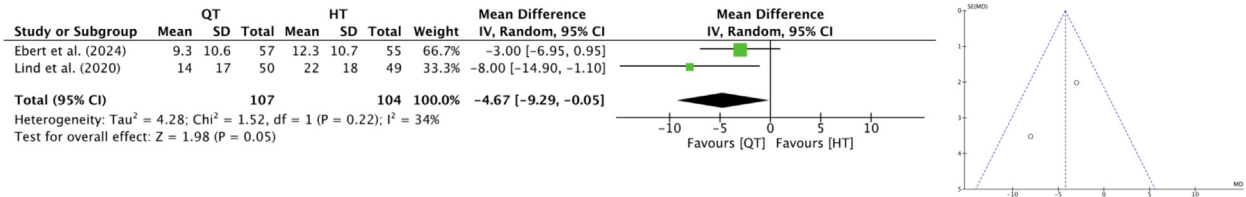


Figure 4. Meta-analysis of post-operative donor site morbidity scores at final follow-up with an associated forest plot.

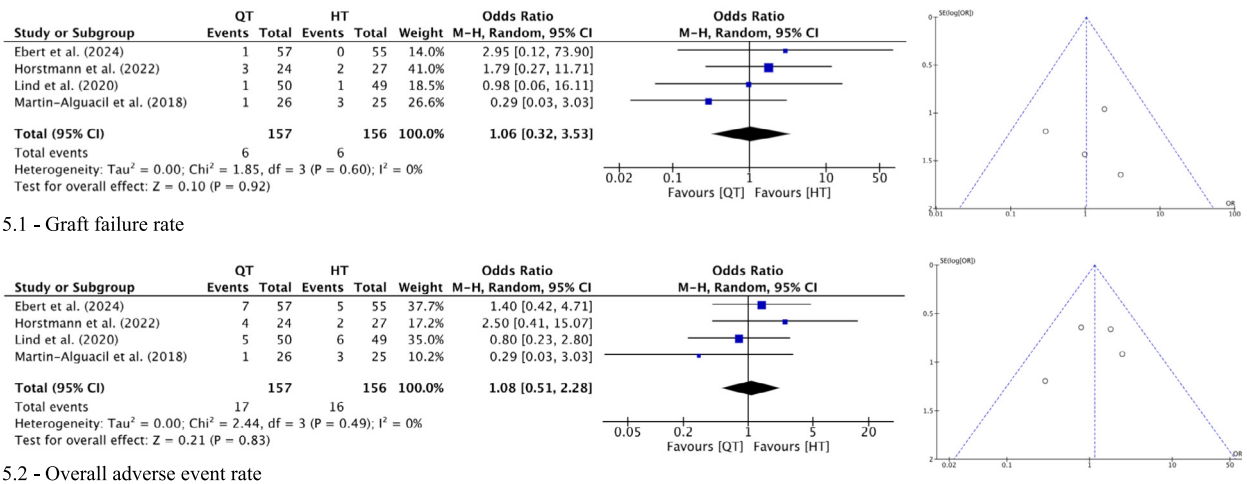


Figure 5. Meta-analysis of post-operative complications: 1) Graft failure rate and 2) Overall adverse event rate at final follow-up with associated forest plots.

3.7. Secondary outcomes: Return to work and return to sport

Only one study reported time taken to return to work and to return to sport [28]. The mean number of days taken to return to work was 45.8 ± 42.7 and 75.2 ± 41.9 in the QT and HT groups, respectively (1 study, n = 51; p = 0.16). Return to sport took 95.2 ± 45.5 days in the QT group and 82.1 ± 45.6 in the HT group, on average (1 study, n = 51; p = 0.62).

3.8. Quality assessment

Risk of bias was assessed as per the RoB 2.0 tool (Figure 6) [24]. All studies were deemed to be at high risk of performance bias as the surgeons needed to be aware of the graft choice intra-operatively. Three articles noted that patients were aware of graft choice [27,29,30], two of which were part of the same trial [29,30]. The other four articles did not report on whether the

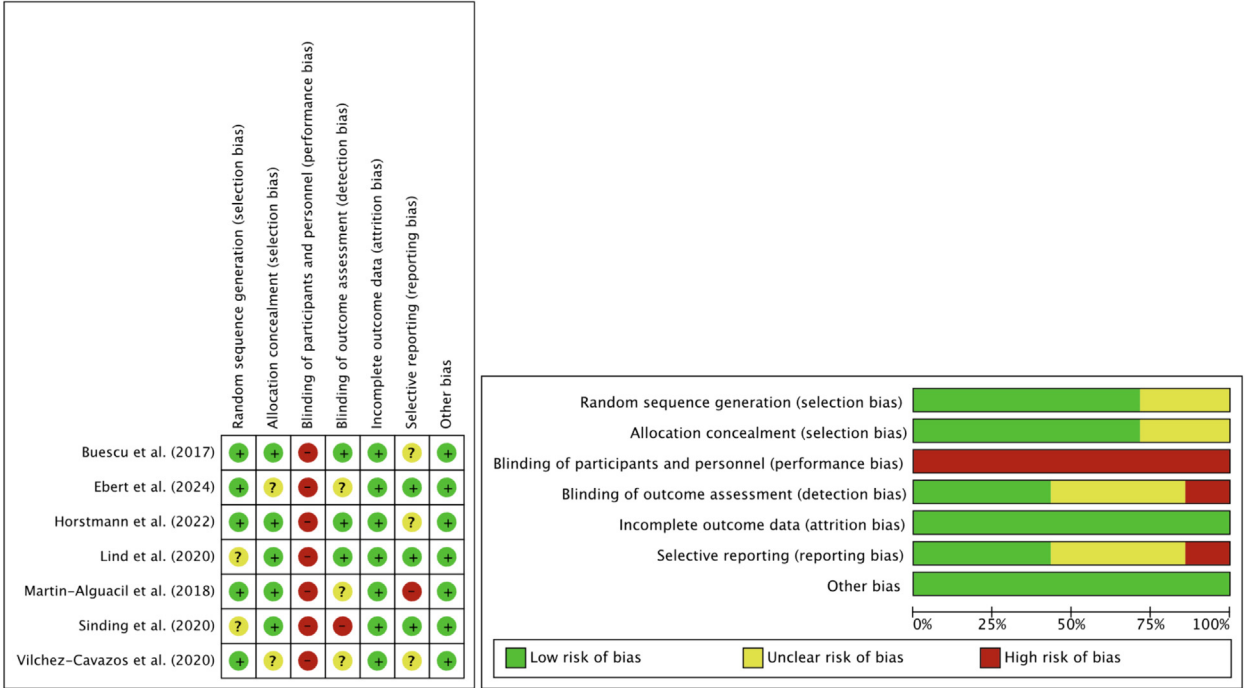


Figure 6. Risk of bias for all included studies with each item presented as percentages.

patient was blinded regarding graft choice [28,31–33]. One article was at high risk of detection bias as evaluators were not blinded due to the procedure-specific scar [30]. Another article was classified as high risk of reporting bias as VAS was not reported despite this being an outcome of interest in the study protocol [32]. It was not possible to determine the risk of reporting bias for three studies as their protocols were not available for review [28,31,33].

4. Discussion

Overall, this systematic review and *meta-analysis* evaluated the outcomes between QT and HT autografts for primary ACLR. There were no significant differences in PROMs, graft failure rates or overall adverse events. Interestingly, there was lower donor site morbidity with the QT autograft as per the *meta-analysis*, which, although promising, warrants further exploration as the MCID for this outcome is not known and the donor site morbidity questionnaire has not yet been validated for use with the QT autograft.

There was no significant difference in the IKDC score between QT and HT groups post-ACLR based on moderate-quality evidence with low heterogeneity. This is in keeping with the most recent review by Tan et al., which also found no difference in IKDC between the groups (MD 0.48 in favour of QT, 95% CI -0.84 to 1.80; 8 studies, 597 patients; $p = 0.48$; $I^2 = 0$) [11]. There were no differences in Lysholm scores or Tegner activity scales as per the *meta-analysis*, which is consistent with the review by Tan et al. [11]. The other PROMs included in this review were not deemed suitable for *meta-analysis* due to the variability in reporting and dearth of data.

Despite equivalent PROMs, this *meta-analysis* identified lower donor site morbidity scores for QT at 24 months based on two studies [27,29], which is a new, significant finding that was not reported in the review by Tan et al. [11], or in a network *meta-analysis* of RCTs by Kunze et al. [34]. It is unclear whether the difference is clinically relevant as the MCID has not been reported for this outcome, but this finding may be important to patients who are concerned regarding donor site symptoms. This finding supports the hypotheses posited by early proponents of QT grafts, who argued for their superiority in terms of donor site morbidity [35,36]. The QT autograft has grown in popularity likely due to its biomechanical properties, such as significantly higher load to failure and greater stiffness [37], as well as an elastic modulus that is similar to the native ACL, which is theorised to be advantageous [35,37,38]. Another potential advantage is the ability to harvest the graft with or without a femoral bone block [39], while having potentially lower donor site morbidity [40] and lack of anterior knee pain, which is associated with BTB grafts as per previous reviews [41,42]. Both studies [27,29] reporting on donor site morbidity scores utilised a questionnaire that has only been validated for HT and BTB grafts [43]. Lind et al. [29] modified the questionnaire such that questions focused on the QT donor site instead of the BTB donor site [43]. While the tool may be valid for the QT graft given that the 16 questions in the questionnaire focus on an unspecified donor site and related symptoms post-ACLR, this may still require further research to ensure validity and reliability.

While none of the studies in this review reported quadriceps or hamstring atrophy, quadriceps atrophy after harvesting a QT autograft can be particularly concerning due to its impact on postoperative recovery and long-term knee function [44]. There are also other risks with QT, such as iatrogenic patellar fracture and inadequate graft length [35], while the risks of HT harvesting include sensory deficits and transient flexion strength deficits [45]. The decision-making process for graft selection might, therefore, hinge on the trade-offs between the known issues with harvesting both grafts and the possibility of lower donor site morbidity with QT grafts. One of the potential challenges of the QT autograft is the learning curve associated with introducing the technique [35,46]. The QT autograft is associated with higher revision rates in low-volume centres compared to high-volume centres, indicating the importance of overcoming the learning curve when adding this technique to the armamentarium for ACLR or when designing a trial [47]. Further comparisons with longer follow-up are needed to identify if the donor site morbidity persists long-term.

Based on the low- to moderate-quality evidence in this review, along with comparable graft failure rates and adverse events between QT and HT autografts, the QT autograft appears to be a safe choice for primary ACLR. This is in keeping with the review by Tan et al. that included registry data and reported comparable graft failure rates (OR 1.07; 95% CI 0.59 to 1.92; 7 studies, 27,691 patients; $p = 0.830$; $I^2 = 65\%$) [11]. These findings may reassure surgeons and patients of the safety of the QT autograft while recruiting patients into future trials.

The disparate data on objective strength scores and the lack of data on the time taken to return to work or to sport represent potential avenues for further research. Tan et al. [11] found that HT autografts were associated with greater limb symmetry indices (LSIs) for knee extensor strength based on a *meta-analysis* of one RCT and two retrospective cohort studies [30,48,49]. Although only one study reported the time taken to return to work and sport [28], the trend in favour of QT suggests that patients may benefit from shorter recover periods. This may be important to patients whose livelihoods depend on returning to work or to sport. Future RCTs could explore these outcomes further to enable a comparative *meta-analysis*, guide rehabilitation protocols and aid patient counselling by considering the specific requirements of respective sports or activities.

There are a few potential reasons that could explain why no differences other than donor site morbidity scores were identified in this review: a lack of RCTs with comparable standardised outcomes; small sample sizes in the included studies, which could limit the statistical power needed to detect meaningful differences between the two autografts; and a maximum follow-up time of 24 months when the time to return to sport can range from 2 to 24 months post-ACLR (mean 7.3 months) [50]. There was marked variability in reporting outcome measures across the included studies, indicating the

need for standardised reporting outcome measures in ACLR trials. This is in keeping with findings from Berk et al., which found that there was “marked heterogeneity and inconsistency” in the PROMs used across 510 studies on ACLR [51]. Svantesson et al. conducted a consensus meeting and recommended that studies should include: early adverse events; PROMs with IKDC recommended as the knee-related outcome measure for ACL treatment; ACL failure rates; and clinical examination measures [52]. Currently, the UK NLR utilises KOOS, IKDC, Euroqol (EQ-5D) and the Tegner activity scale as these enable comparison with other registries [5]. Future studies may benefit from standardised reporting outcome measures to aid in direct comparisons between studies and for pooling data in *meta*-analyses.

At the time of writing, this is the only review to compare QT and HT autografts with a sole focus on RCTs, which was performed to focus on high-quality studies and decrease the likelihood of a skewed *meta*-analysis. The most recent systematic review that compared QT with HT autografts was by Tan et al. in 2022 with a search until 2020 and a *meta*-analysis that showed high heterogeneity, likely due to the pooling of results from RCTs ($n = 6$) and observational studies ($n = 14$) [11]. Two additional RCTs were included in this review [27,28] and there was less heterogeneity (I^2 range 0–56%) within the present *meta*-analyses compared to the previous review (I^2 range 0–96%) [11].

An older review by Mouarbes et al. in 2019 found that the QT autograft led to similar clinical outcomes, functional status and graft survival when compared with BTB and HT autografts with potentially less harvest site pain, however, the results were limited by the inclusion of non-randomised studies and the heterogeneity of studies that underwent *meta*-analysis (27 studies, $n = 2,856$) [41]. This was similar to the narrative reviews by Hurley et al. in 2018, Slone et al. in 2015 and Mulford et al. in 2013, however, these reviews did not include a *meta*-analysis [10,53,54]. A previous network *meta*-analysis of RCTs that investigated donor site morbidity found that the QT autograft was associated with a non-statistically significant reduction in donor-site morbidity compared with the HT autograft, however, both QT and HT autografts were associated with a significant reduction in morbidity compared to the BTB autograft (21 studies, $n = 1,726$) [34]. This is also similar to previous reviews that have compared donor site morbidity between QT and BTB [40,41]. Dai et al. found comparable graft survival, clinical function and pain scores among BTB, HT and QT autografts with significantly lower donor site morbidity scores in the QT group, however, this included registry data (24 studies, $n = 19,584$) [40]. A review by Ajrawat et al. compared these three autografts and found no differences in ACLR revision rates, knee stability and PROMs, however, there was significantly lower anterior knee pain associated with the QT autograft when compared with BTB (10 studies, $n = 1,398$) [42].

The strengths of this review include prospective registration of the review protocol, an up-to-date literature search, the inclusion of a broad range of clinically relevant outcomes and a *meta*-analysis of RCTs. The major limitations include: the small number of eligible RCTs with a pooled sample that is unlikely to be large enough to identify relatively marginal differences in outcomes; the overall low- to moderate-quality evidence; heterogenous outcome reporting; and a maximum follow-up of 24 months. Additionally, this review only focused on two autografts that were investigated in Australian, European and South American institutions, which may limit its generalisability to other populations.

Given the increasing frequency of QT autografts in primary ACLR [6], more robust RCTs with standardised reporting outcome measures and longer follow-up may be needed to identify if a given autograft is superior and, if so, in which specific patient group. For instance, the SQuASH trial (NCT03896464), which compares QT and HT autografts in the paediatric population, is ongoing and its results are awaited [55]. Future trials could also compare different autografts with or without a lateral extra-articular tenodesis to identify indications and contraindications for this adjunct alongside primary ACLR [56–58]. Another factor to consider when designing a trial that compares two autografts is the degree of pre-operative and post-operative rehabilitation provided to patients as this may have a positive impact on PROMs prior to and after the operation [59].

All included studies were at high risk for performance bias and the overall quality of evidence of the outcomes that underwent *meta*-analysis ranged from “low” to “moderate,” which may impact the interpretation of the results. It is unlikely to be able to develop a trial in which the surgeon is blinded from the graft choice, but it may be possible to blind the patient and assessors via sham incisions, for example [60].

Long-term follow-up may also be required to ascertain whether there is a potential difference in donor site morbidity scores, objective strength measures and in graft rupture over time; a review by Wright et al. found that ACL graft rupture rate ranged from 1.8% to 10.4% at 5-year follow-up or longer [61]. By extending the follow-up period, the incidence of post-traumatic osteoarthritis can also be reported and compared [62].

5. Conclusion

Overall, there were no differences in IKDC scores, Lysholm scores, Tegner activity scales, graft failure rates and overall adverse events between the two autografts in primary ACLR as per this *meta*-analysis of RCTs based on low- to moderate-quality evidence. There may possibly be lower donor site morbidity with QT, but the evidence is not sufficient to draw definitive conclusions.

Ethical approval

This systematic review and meta-analysis complies with ethical standards and did not require ethical approval.

Funding

The authors did not receive any funding or financial support for writing this manuscript.

Level of evidence

Level I, meta-analysis of randomised controlled trials.

Registration and protocol

The review protocol was prospective registered on PROSPERO, the International Prospective Register of Systematic Reviews (registration ID CRD42023427339).

CRedit authorship contribution statement

Siddarth Raj: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ali Ridha:** Writing – review & editing, Validation, Investigation, Formal analysis, Data curation. **Henry K.C. Searle:** Writing – review & editing, Validation, Investigation, Formal analysis, Data curation. **Chetan Khatri:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Imran Ahmed:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Andrew Metcalfe:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Nicholas Smith:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Data availability

All of the processed data have been included in the manuscript, tables and figures. The study protocol has been included as an appendix. Any further data are available upon request to the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.knee.2024.07.002>.

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